

Java **LinkedList** Implementation Using Collection

What is **LinkedList** in Java?

LinkedList is a **doubly-linked list** implementation of the **List** and **Deque** interfaces in Java.

It allows **duplicate elements**, maintains **insertion order**, and is ideal for frequent **insertions/deletions**.

```
LinkedList<Type> list = new LinkedList<>();
```

Important Methods of **LinkedList**

Here's a structured table of **methods**, grouped by their use:

Insertion Methods

Method	Description
<code>add(E e)</code>	Add element to end
<code>add(int index, E e)</code>	Add element at index
<code>addFirst(E e)</code>	Add element at start
<code>addLast(E e)</code>	Add element at end (same as <code>add()</code>)
<code>offer(E e)</code>	Add element to queue (end)
<code>offerFirst(E e)</code>	Add at start (queue style)
<code>offerLast(E e)</code>	Add at end (queue style)

Deletion Methods

Method	Description
<code>remove()</code>	Remove first element
<code>remove(int index)</code>	Remove element at index
<code>remove(Object o)</code>	Remove first occurrence of object
<code>removeFirst()</code>	Remove first element
<code>removeLast()</code>	Remove last element

<code>poll()</code>	Retrieve and remove first element
<code>pollFirst()</code>	Remove and return first
<code>pollLast()</code>	Remove and return last
<code>clear()</code>	Removes all elements

Access / Search Methods

Method	Description
<code>get(int index)</code>	Get element at index
<code>getFirst()</code>	First element
<code>getLast()</code>	Last element
<code>peek()</code>	First element (null if empty)
<code>peekFirst()</code>	First element
<code>peekLast()</code>	Last element
<code>contains(Object o)</code>	Returns <code>true</code> if present
<code>indexOf(Object o)</code>	Index of first occurrence
<code>lastIndexOf(Object o)</code>	Index of last occurrence
<code>size()</code>	Returns size of list

Utility Methods

Method	Description
<code>isEmpty()</code>	Check if list is empty
<code>toArray()</code>	Convert to array
<code>iterator()</code>	Return iterator
<code>listIterator()</code>	Bi-directional iterator
<code>descendingIterator()</code>	Reverse iterator